# Maureen Okoro

Data Scientist | Health Data and Machine Learning

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## ABOUT

MSc Data Science student at the University of South Wales with a First Class degree in Pharmacy. I work mainly with health and public data, using Python, SQL, SAS and Power BI to clean and validate data, build statistical and machine learning models, and present findings clearly to non-technical audiences. Pharmacy, where accuracy carries real consequences, gave me strong attention to detail, and I want to use data and machine learning to improve health decisions.

## EDUCATION

**MSc Data Science — University of South Wales** Feb 2026 – Expected Sept 2027
Relevant modules: Applied Statistics (SAS), Big Data Engineering

**BSc Pharmacy — Chukwuemeka Odumegwu Ojukwu University** Nov 2018 – Oct 2023
First Class Honours (Distinction) | CGPA: 4.80 / 5.00

## KEY SKILLS

**Programming:** Python (pandas, NumPy, scikit-learn), SQL (SQL Server), SAS, Dash Plotly, Streamlit, FastAPI
**Machine Learning & Statistics:** Regression, classification, anomaly detection, time series forecasting (Prophet, ARIMA), hypothesis testing
**Data Quality:** Data cleaning, validation, data quality checks, investigating discrepancies, documentation
**Data & Reporting:** Power BI, Excel (Pivot Tables, Power Query), PowerPoint, KPI reporting
**Cloud & Tools:** AWS (S3, EC2, DynamoDB), MongoDB, Jupyter, GitHub

## EXPERIENCE

**Student Navigator and IT Ambassador — University of South Wales** Sept 2026 – Present
*Pontypridd, Wales*

- Support 40 or more students per shift through enrolment and IT setup, explaining systems clearly to non-technical users.

**Healthcare Data Analytics Extern** Sept 2025 – Nov 2025
*Extern.org / Trubridge Healthcare — Remote*

- Performed data quality checks, cleaning and validation on a CDC maternal health dataset spanning 51 U.S. jurisdictions within a reproducible Python pipeline, quantifying a 25.3× mortality disparity between states.
- Built a logistic regression model (0.80 AUC) to identify the key risk factors driving maternal mortality; geographic region explained 20.5% of outcome variance.
- Built an interactive Power BI dashboard and presented findings to non-technical stakeholders.

**Pharmacy Intern — Alpha Pharmacy** Nov 2024 – Nov 2025
*Lagos, Nigeria*

- Verified prescriptions and kept accurate patient and medication records with close attention to detail.
- Supported stock control and documentation audits, checking records against physical stock.

**Data Scientist Intern — DataVerse Africa** Jul 2025 – Sept 2025
*Remote*

- Developed automated Dash Plotly dashboards on trade and customs data, detecting under-taxed goods and delivering revenue policy recommendations.

## PROJECTS

**Road Speed and Injury Severity in Wales: Statistical Analysis — Link** 2026

- Group project in SAS on 2025 STATS19 collision data: found that 46% of collisions on 60 mph roads killed or seriously injured someone, against 35% across all roads, using correlation ($r = 0.59$, $p = 0.004$), paired t and Mann-Whitney U tests with robustness checks.

**Air Pollution and Respiratory Disease in Lagos: Forecasting — Link** 2025

- Linked pollution and hospital respiratory case data (2021 to 2023); regression models predicted poorly ($R^2$ about 0.04), while Prophet and ARIMA captured yearly seasonality and gave reliable short term forecasts.

**FraudWatch Africa: Mobile Money Fraud Detection — Link** 2025

- Built a fraud detection system using Isolation Forest on 10,000 transactions, finding that 77% of anomalies came from agent transactions; served predictions through a FastAPI endpoint and a Streamlit dashboard.

## CERTIFICATIONS & LEARNING

**AWS Academy:** Data Engineering (in progress), Cloud Foundations, ML Foundations, Generative AI Foundations | **DataCamp:** Data Scientist & Associate Data Scientist | **Forage:** BCG Data Science, Deloitte Data Analytics